8 Image captioning

- *image captioning* represents the task of describing the content of an image in natural language, i.e., generating captions for an image, thus its name
- it is a fundamental problem in artificial intelligence that connects computer vision and natural language processing
- it mimicks the remarkable human ability to compress huge amounts of visual infomation into descriptive language

8 Image captioning



A female tennis player in action on the court.



A group of young men playing a game of soccer



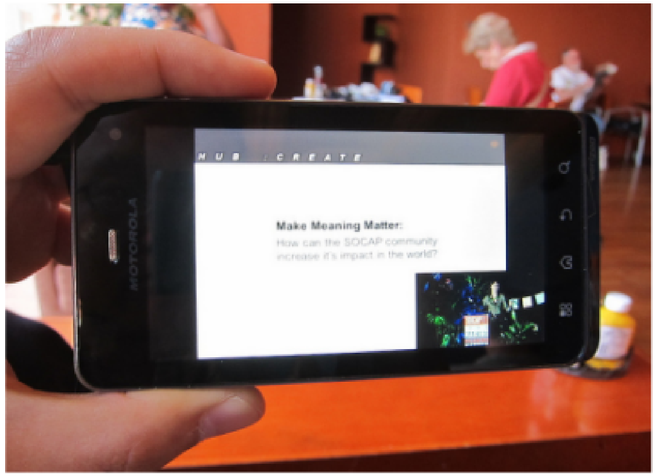
A man riding a wave on top of a surfboard.



A baseball game in progress with the batter up to plate.



A brown bear standing on top of a lush green field.



A person holding a cell phone in their hand.

8 Image captioning

- the task of free-form and open-ended *visual question answering (VQA)*: given an image and a natural language question about the image, the task is to provide an accurate natural language answer
- mirroring real-world scenarios, such as helping the visually impaired, both the questions and answers are open-ended
- visual questions selectively target different areas of an image, including background details and underlying context
- as a result, a system that succeeds at VQA typically needs a more detailed understanding of the image and complex reasoning than a system producing generic image captions
- nonetheless, many open-ended answers contain only a few words or a closed set of answers that can be provided in a multiple-choice format

8 Image captioning



Q: What endangered animal is featured on the truck?

A: **A bald eagle.**
A: A sparrow.
A: A humming bird.
A: A raven.



Q: Where will the driver go if turning right?

A: **Onto 24 ¾ Rd.**
A: Onto 25 ¾ Rd.
A: Onto 23 ¾ Rd.
A: Onto Main Street.



Q: When was the picture taken?

A: **During a wedding.**
A: During a bar mitzvah.
A: During a funeral.
A: During a Sunday church service.



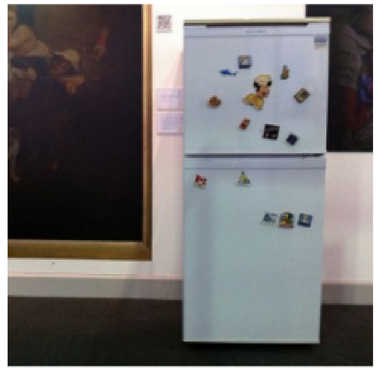
Q: Who is under the umbrella?

A: **Two women.**
A: A child.
A: An old man.
A: A husband and a wife.



Q: Why was the hand of the woman over the left shoulder of the man?

A: **They were together and engaging in affection.**
A: The woman was trying to get the man's attention.
A: The woman was trying to scare the man.
A: The woman was holding on to the man for balance.



Q: How many magnets are on the bottom of the fridge?

A: **5.**
A: 2.
A: 3.
A: 4.

8 Image captioning

- the task of *visual dialog* requires an agent to hold a meaningful dialog with humans in natural, conversational language about visual content
- specifically, given an image, a dialog history, and a question about the image, the agent has to ground the question in the image, infer context from history, and answer the question accurately
- visual dialog is disentangled enough from a specific downstream task so as to serve as a general test of machine intelligence, while being grounded in vision enough to allow objective evaluation of individual responses and benchmark progress

8 Image captioning

